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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The general push-forward statement for tor amplitude. Lemma \ref{lemma-push-tor-amplitude}, cited at proof line 17128. Identifies the result as a special case of a more general statement.
- A module over an algebra of tor dimension at most d has derived base change with cohomology only in degrees minus d through zero. assumption on the tor dimension, used at proof lines 17141-17143. Bounds the degrees in which the base changed test module can have cohomology.
- The spectral sequence computing the cohomology of a total complex. Example \ref{example-tor}, cited at proof lines 17143-17145. Converts the two degree bounds into the shifted bound on the cohomology of the tensor product.

Source reference. Original lemma 066K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7939–7939. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0B66: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14059–14059. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0662: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7921–7921. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a ring map. Assume that B has tor dimension $\leq d$ as an A-module. Let $K^\bullet$ be a complex of B-modules. Let $a, b \in \mathbf{Z}$. If $K^\bullet$ as a complex of B-modules has tor amplitude in $[a, b]$, then $K^\bullet$ as a complex of A-modules has tor amplitude in $[a - d, b]$.*

Proof. This is a special case of Lemma 0B66, but can also be seen directly as follows. Let M be an A-module. Choose a free resolution $F^\bullet \to M$. Then

$$K^\bullet \otimes^{\mathbf{L}}_A M = \mathrm{Tot}(K^\bullet \otimes_A F^\bullet) = \mathrm{Tot}(K^\bullet \otimes_B (F^\bullet \otimes_A B)) = K^\bullet \otimes^{\mathbf{L}}_B (M \otimes^{\mathbf{L}}_A B).$$

By our assumption on B as an A-module we see that $M \otimes^{\mathbf{L}}_A B$ has cohomology only in degrees $-d, -d+1, \ldots, 0$. Because $K^\bullet$ has tor amplitude in $[a, b]$ we see from the spectral sequence in Example 0662 that $K^\bullet \otimes^{\mathbf{L}}_B (M \otimes^{\mathbf{L}}_A B)$ has cohomology only in degrees $[-d + a, b]$ as desired. □